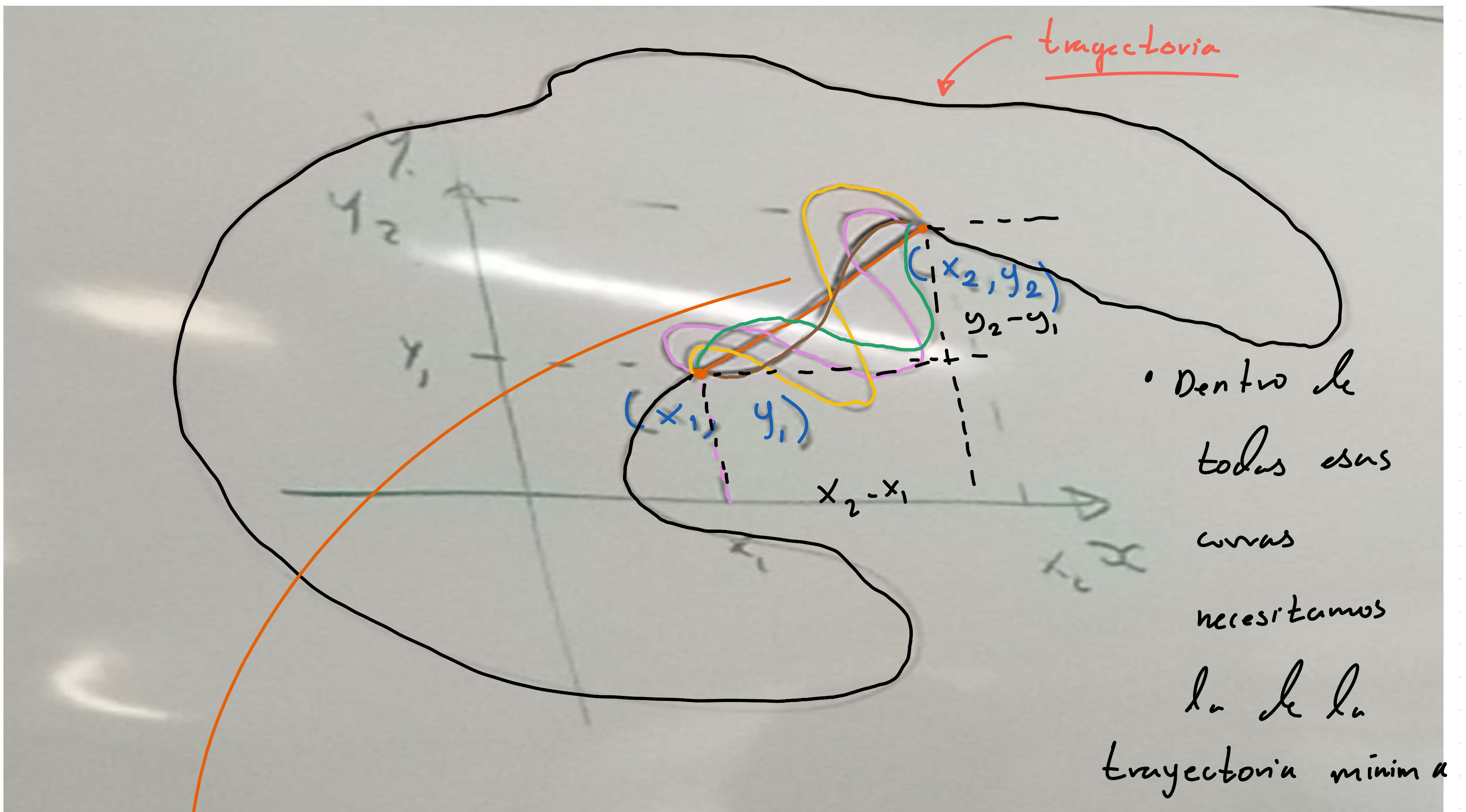
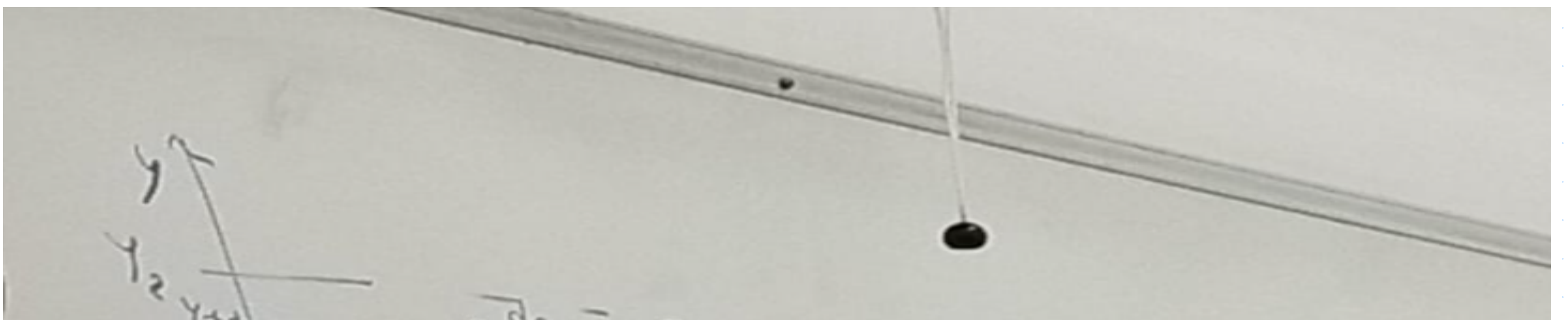


Capítulo II → Principio de Hamilton

- Requerimos saber cómo se define la longitud.
- curvas suaves (está bien definida la recta tangente).

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Curva arbitraria de toda la familia de curvas anteriores:

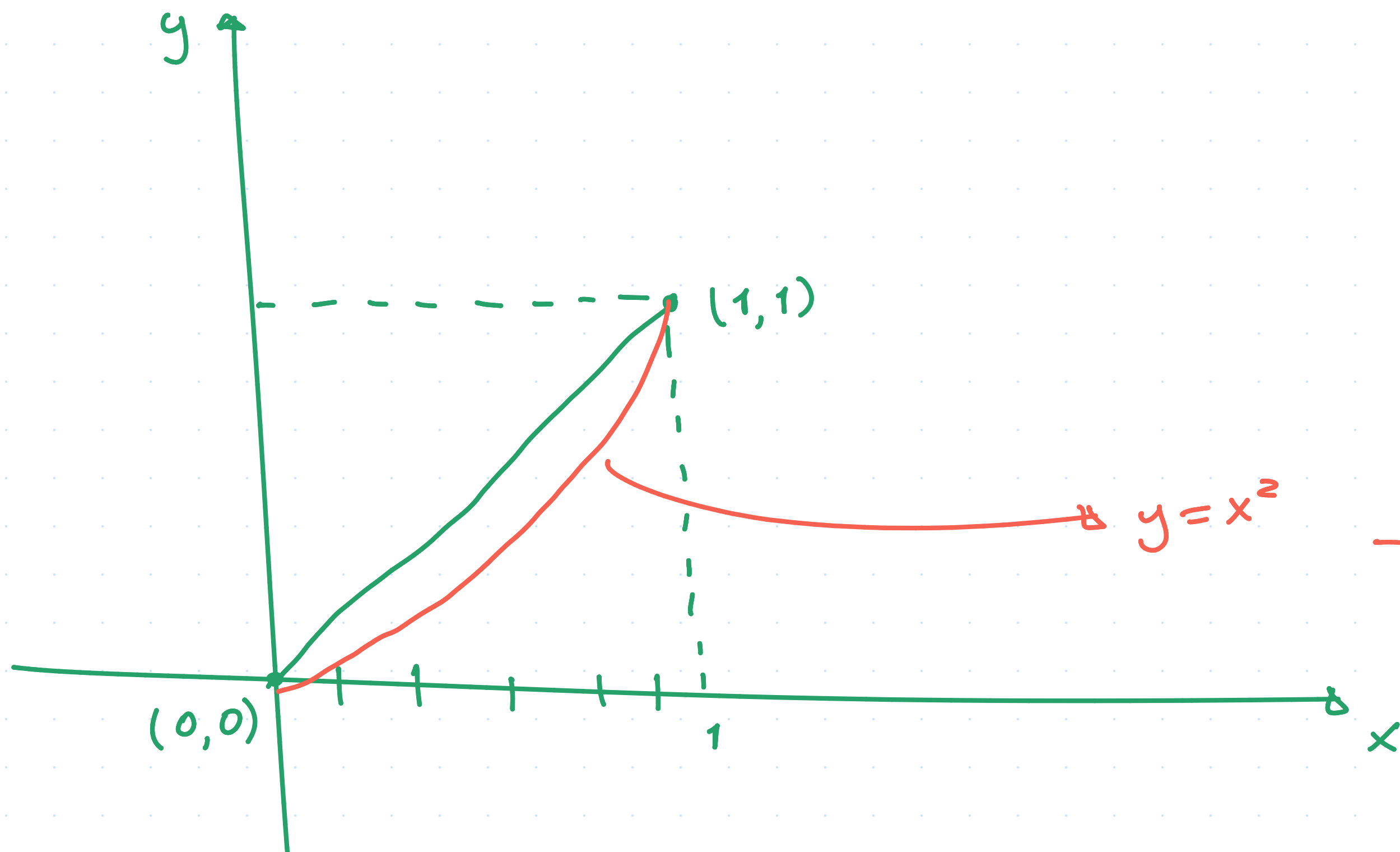


$$S_{1,2} = \int_{x_1}^{x_2} (1 + \dot{y}^2)^{1/2} dx$$

Siempre y cuando

$$y = y(x)$$

y como función de x .



$$y = x$$

$$\frac{dy}{dx} = 1$$

Tenemos

$$S_{12}^{LR} = \int_0^1 \sqrt{1 + 1^2} dx = \sqrt{2} \int_0^1 dx$$

Parábola

$$S_{12}^P = \int_0^1 \sqrt{1 + \left(\frac{dx^2}{dx}\right)^2} dx$$

$$\hookrightarrow S_{12}^{LR} = \sqrt{2} x \Big|_0^1 = \sqrt{2} = 1.41421$$

$$\int_0^1 \sqrt{1 + (2x)^2} dx$$

$$= \int_0^1 \sqrt{1 + 4x^2} dx = \underline{1.47814}$$

¡chequear integral!

$$\Rightarrow S^{\mathbb{R}} \subset S^{\mathbb{P}} \quad \left\{ \begin{array}{l} \cdot \text{ Línea recta es} \\ \text{la trayectoria mínima} \end{array} \right.$$

Tenemos

$$y = y(x)$$

$$y = y(x, \alpha) \quad \} \quad \alpha \rightarrow 0 \quad \text{representan trayectorias infinitesimales}$$

Tenemos

$$S_{12} = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{\partial y}{\partial x} \right)^2} dx$$

Tenemos

$$\delta S_{12}(\alpha) = 0$$

$$\delta S_{12}(\alpha) = \left(\frac{dS_{12}}{d\alpha} \right) \bigg|_{\alpha=0} \delta x$$

Tenemos

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \quad \Rightarrow \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

$$S_{12} = \int_{x_1}^{x_2} \sqrt{1 + \dot{y}^2} dx \Rightarrow \frac{d}{dx} \left(\frac{\partial f}{\partial \dot{y}} \right) - \frac{\partial f}{\partial y} = 0$$

$$f(y, \dot{y}, x) = \sqrt{1 + \dot{y}^2}$$

$$\frac{\partial f}{\partial y} = 0 \Rightarrow \frac{d}{dx} \left(\frac{\partial f}{\partial \dot{y}} \right) = 0$$

cte

De aquí se desprende que:

$$\frac{\partial f}{\partial \dot{y}} = cte ; \quad \frac{\partial f}{\partial \dot{y}} = \frac{\dot{y}}{\sqrt{1 + \dot{y}^2}} \quad 2\dot{y} = \frac{\dot{y}}{\sqrt{1 + \dot{y}^2}}$$

Tenemos

$$\frac{\dot{y}}{(1 + \dot{y}^2)^{3/2}} = cte = c$$

"Elevamos todo al cuadrado"

$$\frac{\dot{y}^2}{1 + \dot{y}^2} = c^2$$

$$\dot{y}^2 = c^2 + c^2 \dot{y}^2$$

$$\vdots$$

$$\dot{y}^2 (1 - c^2) = c^2$$

$$\dot{y}^2 = \frac{c^2}{1 - c^2}$$

$$\dot{y} = \pm \sqrt{\frac{c^2}{1 - c^2}}$$

$-1 \leq c \leq 1$ $\left\{ \begin{array}{l} \cdot \text{ seno} \\ \cdot \text{ coseno} \\ \text{viven entre} \end{array} \right.$

Tenemos (cambio de coordenadas)

$$c = \sin \alpha$$

$$\frac{c^2}{1-c^2} = \frac{\sin^2 \alpha}{1-\sin^2 \alpha} = \frac{\sin^2 \alpha}{\cos^2 \alpha} = \tan^2 \alpha.$$

Tomamos en la func. tangente ya se incorporó " $\pm$ "

$$\dot{y} = \tan \alpha$$

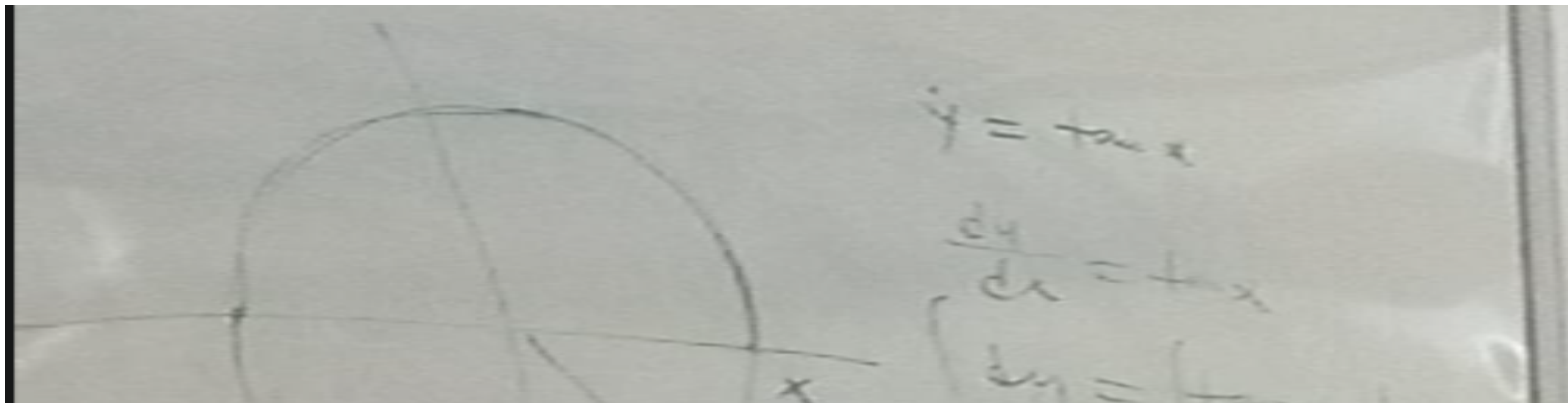
↪ Cambiamos de notación

$$\frac{dy}{dx} = \tan \alpha$$

$$\int dy = \int \tan \alpha \, dx$$

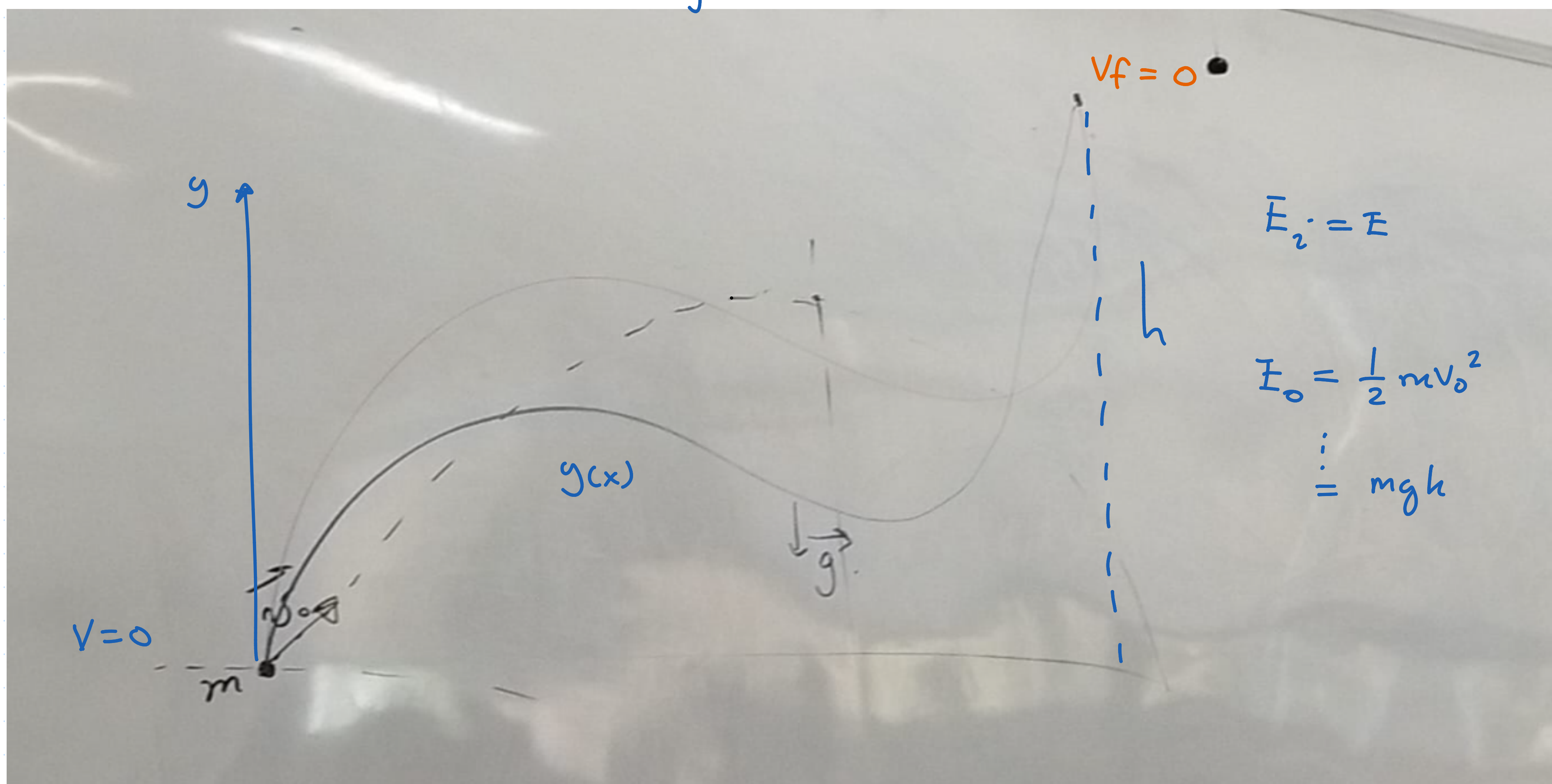
$$y = (\tan \alpha) x + b$$

$$\Rightarrow y = mx + b$$



Ejemplo 2.

calcular el tiempo desde el punto más bajo al más alto.



• Considerar conservación de la energía.

$$\underline{E_i = E}$$

E en un punto arbitrario:

E en un punto cualquiera.

$$E_0 = \frac{1}{2} m v_0^2 = \frac{1}{2} m \underset{\substack{\uparrow \\ \text{depende del tiempo}}}{v^2} + \overbrace{mgy(x)}$$

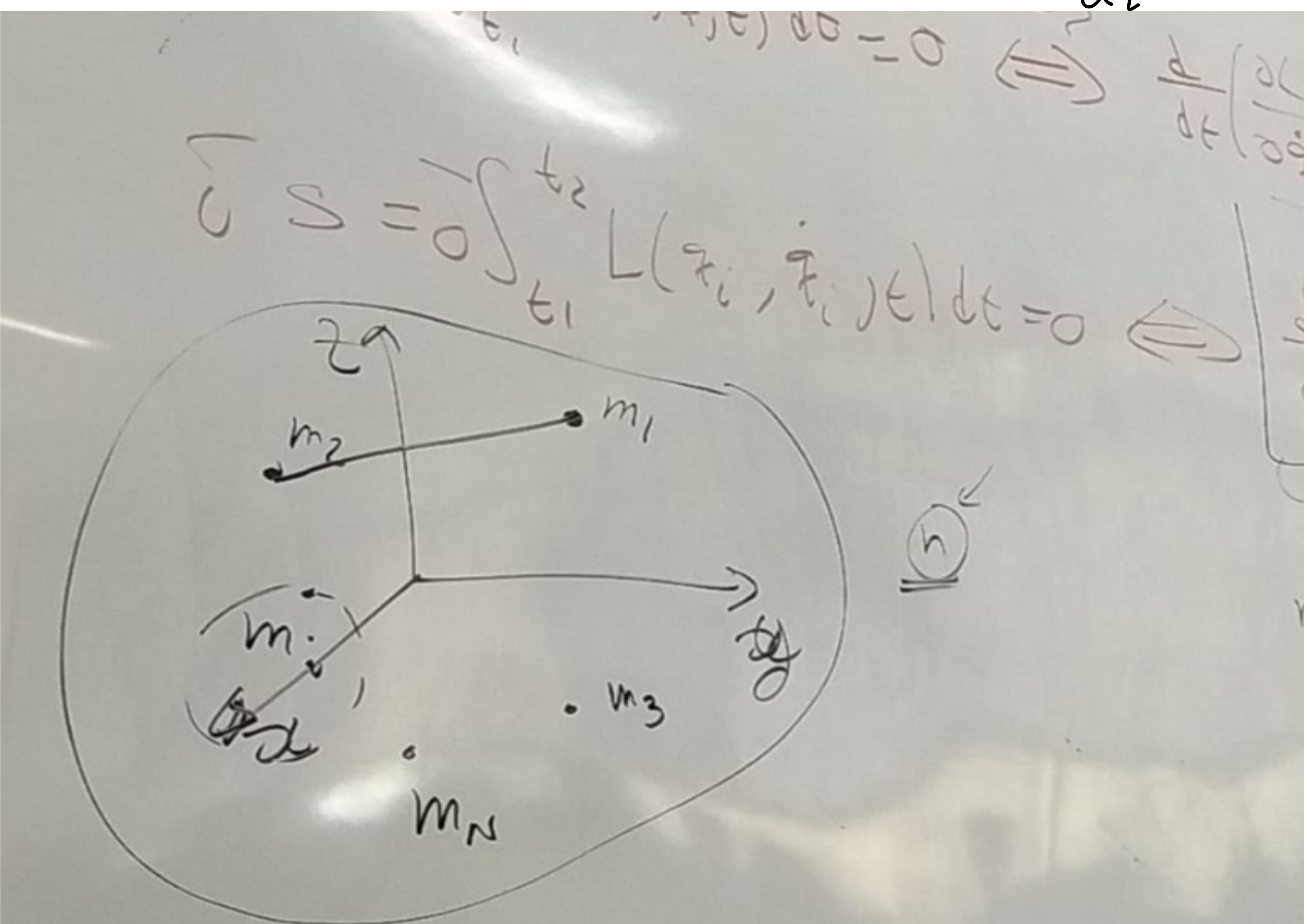
Tenemos...

$$\delta S = \delta \int_{t_1}^{t_2} L(q, \dot{q}, t) dt = 0 \quad \Leftrightarrow \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

$$\delta S = \delta \int_{t_1}^{t_2} L(q_i, \dot{q}_i, t) dt = 0 \quad \Leftrightarrow \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

$\{i\}_{i=1}^n$

$$m_a \frac{d^2 \vec{r}_a}{dt^2} = \vec{F}_a + \vec{T}_{ca}$$



$$\vec{r}_a$$

$$\vec{p}_a = m_a \vec{v}_a$$

$$E_a = \frac{1}{2} m_a \vec{v}_a^2$$

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}$$

$$\frac{d\vec{p}}{dt} = \vec{F} = \vec{0} \Rightarrow \vec{p} = cte$$

